



Report on Task 3

**Title: Rice Variety Classification from Microscopic Images using CNN with Attention:
Model Improvement, Ablation Study and Explainability Analysis**

Course Title: Machine Learning

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1. Introduction

Rice variety identification from microscopic images is an important computer vision application for automated quality assessment, seed purity verification, and precision agriculture. This task focuses on improving the previously developed CNN classification system by introducing attention mechanisms, optimizing training strategies, validating performance statistically, and interpreting model decisions.

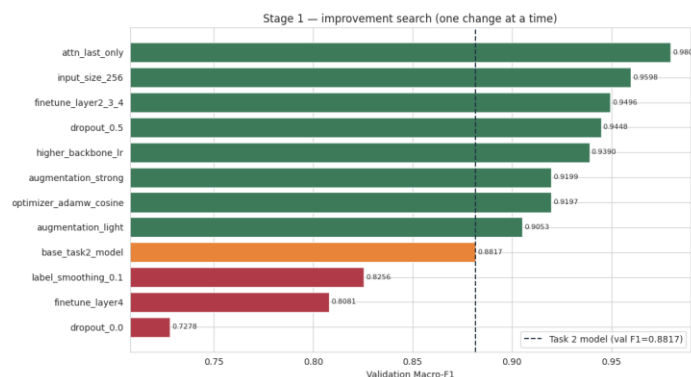
The complete workflow combines performance improvement with explainability. Instead of reporting only classification accuracy, the study investigates why the model improves and whether it focuses on meaningful rice grain characteristics.

2. Baseline and Improved Model Development

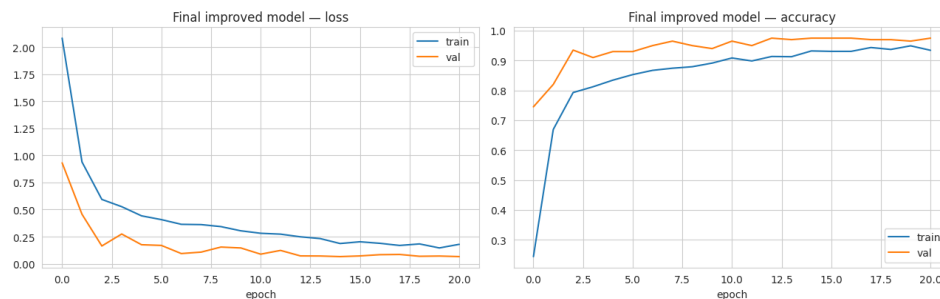
The baseline architecture uses ResNet50 as a deep feature extractor. The improvement framework introduces the Convolutional Block Attention Module (CBAM), which enhances feature learning by assigning importance weights to informative channels and spatial regions.

The final improved configuration was selected after controlled experiments involving fine-tuning depth, attention placement, dropout, optimization strategy, learning rate scheduling, and augmentation strength.

Model improvement experiment output generated from the notebook:



Training and validation loss/accuracy curves for the final improved model, generated from the notebook:

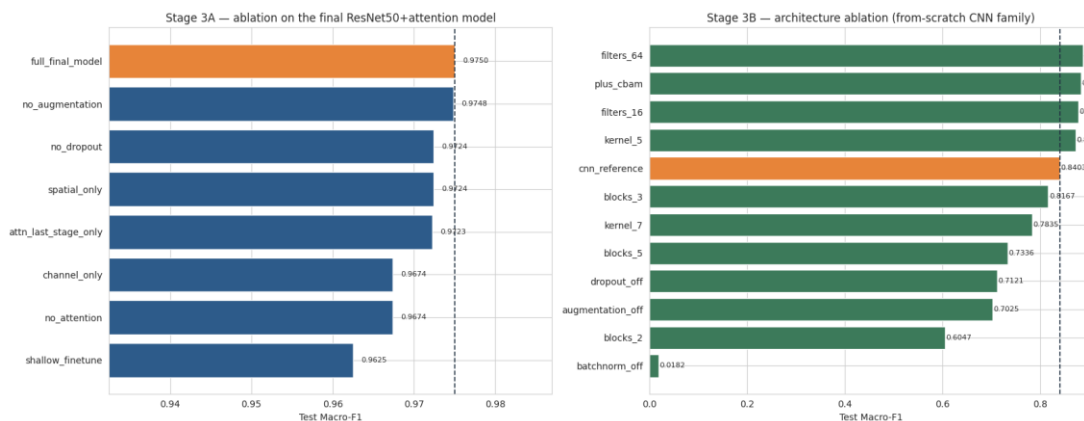


3. Optimization and Ablation Study

Ablation analysis was performed to identify the contribution of individual components. Attention removal, channel-only attention, spatial-only attention, and training strategy modifications were compared with the complete model.

The results demonstrated that the complete CBAM-enhanced model provided the most balanced improvement, confirming that attention-based feature refinement contributes positively to rice variety discrimination.

Ablation comparison output generated from the notebook, showing both the ablation on the final ResNet50+attention model (left) and the architecture ablation on the from-scratch CNN family (right):

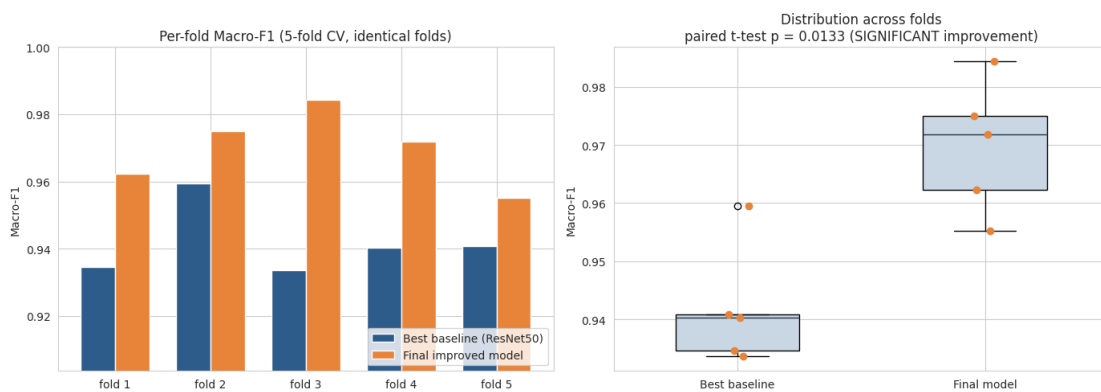


4. Cross Validation and Statistical Validation

To verify reliability, stratified five-fold cross-validation was performed using the training-validation data while keeping the test set independent. Mean Macro-F1 score and variation across folds were calculated.

Statistical tests including paired comparison analysis and prediction-level comparison were used to determine whether the improvement was consistent rather than caused by random variation.

Cross-validation and evaluation results generated from the notebook:

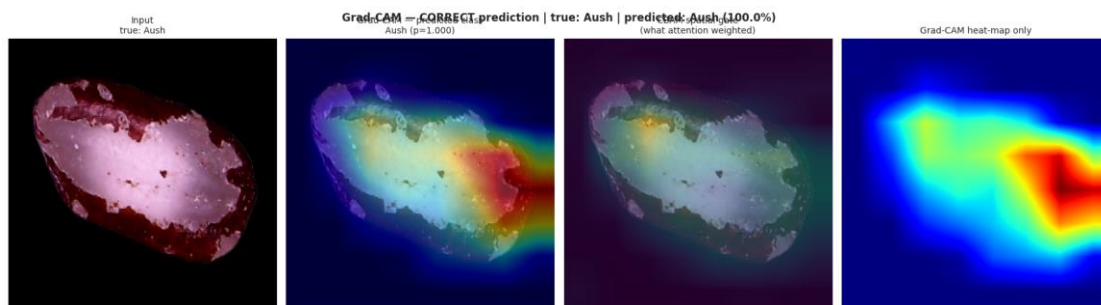


5. Explainability Analysis

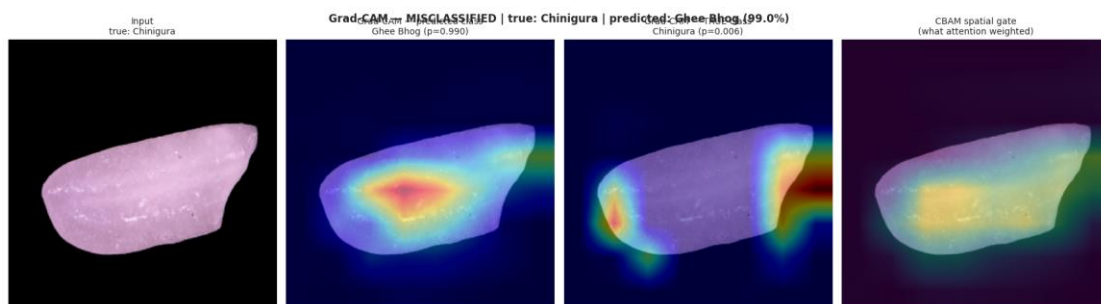
The explainability analysis evaluates whether the model makes decisions based on meaningful image regions. Grad-CAM was applied to visualize areas contributing to predictions, while LIME provided local explanations by analysing image regions through controlled perturbations.

These methods improve model transparency by showing which parts of the rice grain image influence classification decisions.

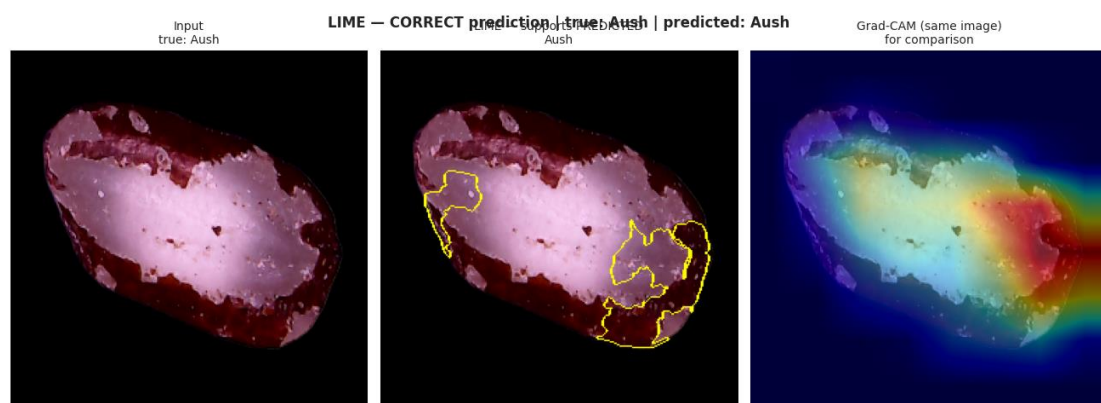
Grad-CAM explanation output generated from the notebook:



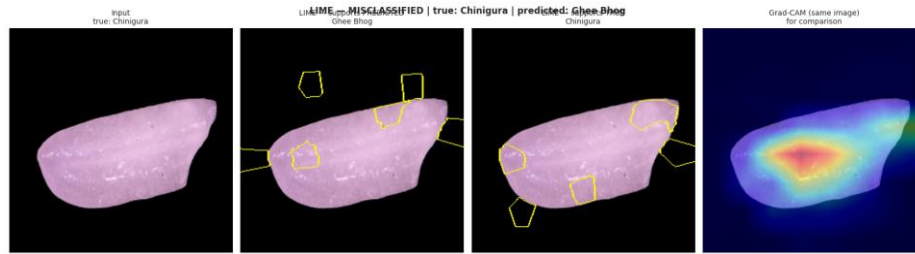
Grad-CAM explanation output for the required misclassified (wrong) prediction, generated from the notebook. The true class is Chinigura, predicted as Ghee Bhog with 99.0% confidence; the third panel shows Grad-CAM computed for the true class instead of the predicted class:



LIME and prediction explanation output generated from the notebook, for the correct prediction (true: Aush, predicted: Aush):



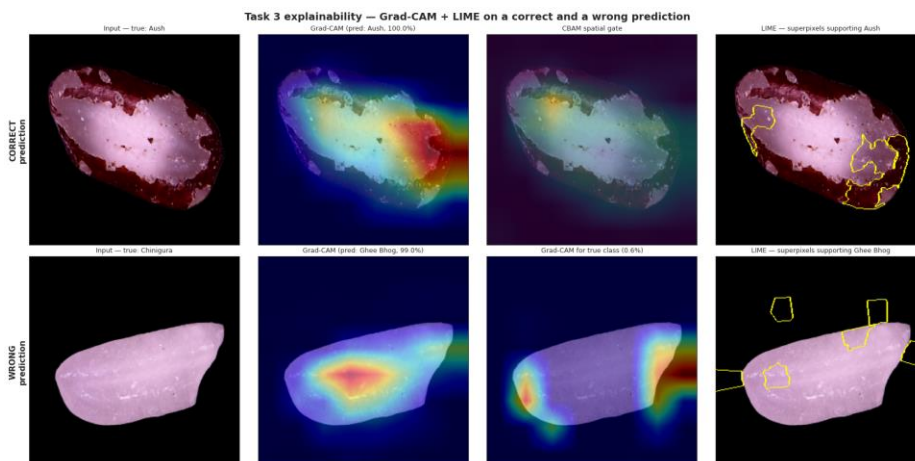
LIME explanation output for the required wrong prediction (true: Chinigura, predicted: Ghee Bhog), generated from the notebook:



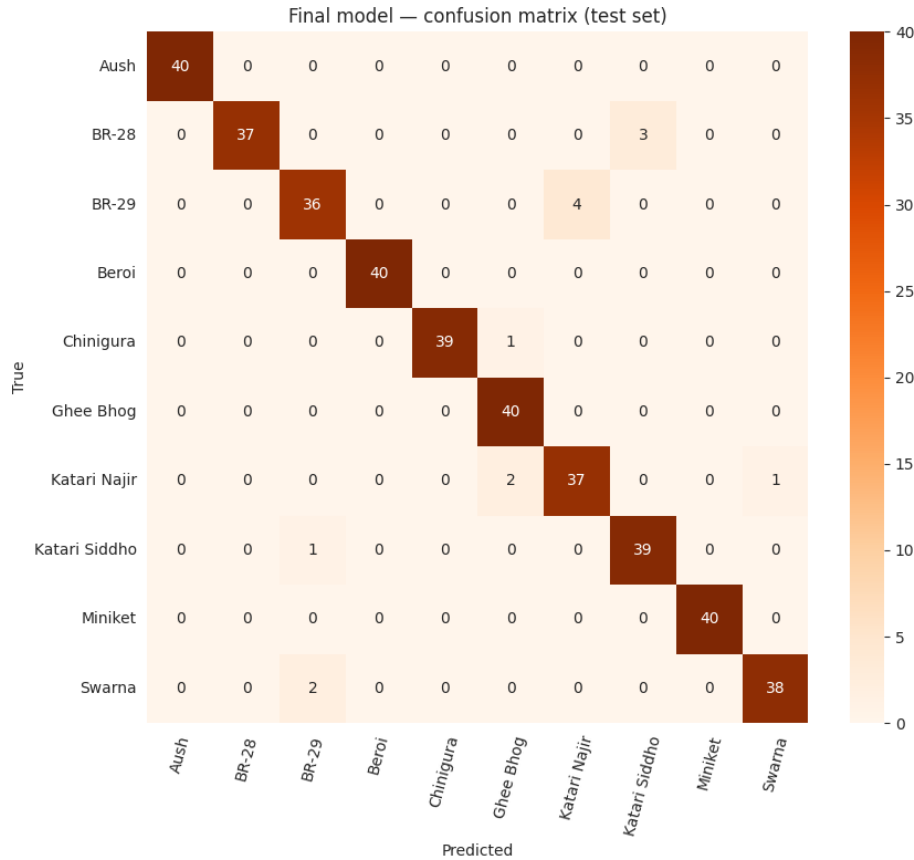
For the correct prediction, Grad-CAM concentrates on the grain body rather than the background, indicating that the decision is driven by rice texture itself; Grad-CAM and LIME show weak-to-moderate agreement ($\text{IoU} = 0.174$ between the LIME superpixel mask and the top-30% Grad-CAM region), which is meaningful corroboration since LIME does not use the model's gradients at all.

For the wrong prediction, the model assigned only 0.6% probability to the true class Chinigura, so this is a confident error rather than a borderline one. Comparing the Grad-CAM map for the predicted class (Ghee Bhog) with the map for the true class (Chinigura) shows the two maps sit in overlapping but not identical regions of the grain, suggesting the model located a plausible region of evidence but misread the fine-grained texture rather than looking at the wrong part of the image entirely. Grad-CAM and LIME again show weak agreement on this image ($\text{IoU} = 0.080$), consistent with the explanation being less stable on a misclassified, harder example.

Combined Grad-CAM and LIME summary panel - correct prediction (top row) versus wrong prediction (bottom row), generated from the notebook:



Confusion matrix on the test set, used for per-class error analysis, generated from the notebook:



The dominant confusion pair is BR-29 being misread as Katari Najir (4 test images), consistent with the fine-grained texture similarity these two varieties share under microscopic imaging. Overall, Grad-CAM and LIME agree that the model's decisions are grounded in the rice grain region rather than background artefacts, and the error analysis indicates the model's mistakes are concentrated among visually similar variety pairs rather than being random.

6. Conclusion

The Task 3 experiments demonstrate that combining CNN optimization, CBAM attention, rigorous ablation analysis, statistical validation, and explainability creates a reliable deep learning framework for rice variety classification. The

final model is not only accurate but also interpretable, allowing better understanding of the visual features used for prediction.